PropRent

Property Rental Management & Tenant Portal Platform

Chapter 3: Requirements Gathering Specification

Version 1.0

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Updated Baseline Scope Summary	8 key stakeholders, 8 core user stories, 20 functional requirements across the 7 core epics (17 Must, 2 Should, 1 Could) and 10 non-functional requirements (30 requirements in total).

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Purpose of this chapter

This chapter establishes the baseline of what PropRent must do (20 functional requirements) and how well it must do it (10 non-functional requirements). It identifies the 8 key stakeholders affected by the system, translates their needs into 8 user stories and 3 use cases, and specifies verifiable requirements that drive design, implementation and testing. The scope has been deliberately focused following supervisor feedback.

1. Stakeholder Analysis

1.1 Classification Approach

Stakeholders are classified using the following definitions:

- **Primary** stakeholders use the system directly and daily; their success defines the success of the product.
- **Secondary** stakeholders interact with the system or its outputs on a recurring basis, or provide services the platform depends on.
- **Tertiary** stakeholders do not use the system but impose constraints (legal, financial, governance) or are indirectly affected by its outcomes.

Influence is rated **High** (can approve, block or fundamentally reshape the product), **Medium** (shapes specific features or integrations) or **Low** (indirect or occasional effect). Interest is rated by how directly the stakeholder is affected by day-to-day platform behaviour.

> **Scope note**
>
> In response to supervisor feedback, the register has been consolidated to **8 key stakeholders**. Technical suppliers (e-signature, notification and hosting services) are treated as integration and infrastructure dependencies in the design chapters rather than as stakeholders, and the regulatory bodies and the academic evaluation committee are merged into one governance group.

1.2 Stakeholder Register

Stakeholder / Role & Classification	Core Business Needs & Pain Points	Level of Influence / Impact
Tenant / Renter *Primary* Individual or household occupying a rented unit; pays rent, reports defects and communicates with management.	**Needs:** one place to pay rent, view the lease and receipts, and track repairs. **Pain points:** missed due dates, opaque payment status, repairs reported by phone with no tracking, lost paperwork.	High Influence / High Interest
Property Manager / Leasing Agent *Primary* Day-to-day operator of one or more properties; lists units, creates leases, monitors collections and triages maintenance.	**Needs:** centralised portfolio view, automated billing, structured work orders, lease templates. **Pain points:** spreadsheet and email workflows, manual rent chasing, duplicated data entry, no audit trail.	High Influence / High Interest
Property Owner / Landlord *Primary* Legal owner of one or more properties; receives rental income and bears the financial risk.	**Needs:** transparent income, occupancy and expense reporting; control over approvals above spending thresholds. **Pain points:** delayed statements, vacancy losses, unauthorised or overpriced repairs.	High Influence / High Interest
Platform Administrator *Primary* Technical and operational administrator of the deployment; manages users, roles, configuration and audit logs.	**Needs:** secure user and role administration, configuration without code changes, audit logs. **Pain points:** manual account recovery, unclear permission boundaries, difficulty tracing incidents.	High Influence / High Interest
Prospective Tenant / Applicant *Secondary* Person browsing listings and submitting enquiries; becomes a Tenant on lease execution.	**Needs:** accurate searchable listings with photos, easy enquiry and clear replies. **Pain points:** outdated listings, slow replies, repeated form filling.	Medium Influence / High Interest
Maintenance Contractor / Vendor *Secondary* Plumbers, electricians and technicians performing repair work on assignment.	**Needs:** clear job descriptions with photos, access instructions, ability to update status and record final cost. **Pain points:** vague tickets, repeated site visits, delayed approvals and payment.	Medium Influence / High Interest
Payment Gateway Provider *Secondary* External PCI-DSS Level 1 provider processing card, bank-debit and wallet payments, tokenisation and webhooks.	**Needs:** correct API usage, idempotent requests, secure webhook handling. **Pain points:** duplicate charges from retries, failed webhooks, chargeback handling.	High Influence / Low-to-Medium Interest *(Technical & compliance dependency)*
Regulatory & Legal Compliance Bodies / Academic Evaluation Committee *Tertiary* Housing, data-protection, consumer-protection and tax authorities, together with the DEPI committee that evaluates scope, rigour and delivery.	**Needs:** lawful handling of tenant data, record retention and notice periods; well-bounded scope and traceable, measurable requirements. **Pain points:** non-compliant lease clauses, scope creep, unmeasurable requirements.	High Influence / Lower Interest *(Governance & compliance)*

1.3 Influence / Interest Summary

Quadrant	Stakeholders	Engagement Strategy
High influence, high interest	Tenant / Renter, Property Manager / Leasing Agent, Property Owner / Landlord, Platform Administrator	Manage closely: interviews, prototype reviews, UAT sign-off.
High influence, lower interest	Payment Gateway Provider; Regulatory & Legal Compliance Bodies / Academic Evaluation Committee	Keep satisfied: compliance checklists, integration reviews, milestone reports.
Medium influence, high interest	Prospective Tenant / Applicant, Maintenance Contractor / Vendor	Keep informed: usability sessions, targeted feedback on listing and work-order flows.
Lower influence, lower interest	None after consolidation to eight stakeholders	Monitor: periodic review of integration needs.

Key insight

Tenants and Property Managers are the two groups whose daily workflows intersect on every core process (payments, maintenance, leases). Any requirement that improves one group's experience at the expense of the other should be escalated for explicit trade-off analysis.

2. User Stories & Use Cases

2.1 User Stories

Eight high-priority stories, one per key stakeholder, capture the core value of the platform. Each follows the format *As a [role], I want to [action], so that [benefit]* and carries three Given-When-Then acceptance criteria. Story IDs are referenced by the functional requirements in Section 3.

US-01: Pay Rent and Report Repairs (Tenant / Renter)

As a tenant, **I want to** pay my rent online and report repair issues with photos or video from one mobile-friendly portal, **so that** I avoid late fees, keep proof of payment and never need to chase anyone by phone.

Acceptance Criteria

1. **Given** I am logged in and have an invoice with status Due, **when** I select a saved payment method and confirm, **then** the system charges the exact invoice amount, marks the invoice Paid and emails a PDF receipt with a confirmation reference.
2. **Given** my payment is declined by the gateway, **when** the response is received, **then** the invoice remains Due, I see a plain-language reason and I am offered another payment method.
3. **Given** I am logged in with an active lease, **when** I submit a request with category, description, urgency and up to 5 photos or videos, **then** a ticket with a unique ID and status Submitted is created and I am notified of every later status change.

Traces to: FR-FIN-01, FR-FIN-02, FR-MAINT-01, FR-MAINT-02, FR-COMM-01, FR-RPT-03 | UC-01, UC-02

US-02: Publish Units, Sign Leases and Monitor Operations (Property Manager / Leasing Agent)

As a property manager, **I want to** publish units, send leases for electronic signature and monitor collections and open tickets on one dashboard, **so that** vacant units are marketed quickly and tenant onboarding needs no paper or in-person meetings.

Acceptance Criteria

1. **Given** I have manager permissions and a unit with all mandatory fields (address, type, rent, deposit, availability date, at least one photo), **when** I click Publish, **then** the unit appears in public search and records who published it and when; if a mandatory field is missing, publication is blocked and each missing field is highlighted.
2. **Given** an approved applicant and a unit, **when** I generate a lease from a template, send it for signature and every signer has signed, **then** the lease status becomes Active and the signed PDF with its audit certificate is stored.
3. **Given** invoices and tickets exist, **when** I open my dashboard, **then** I see occupancy rate, rent collected versus expected, overdue balances, open tickets and upcoming lease expiries.

Traces to: FR-PROP-01, FR-PROP-02, FR-LEASE-01, FR-LEASE-02, FR-LEASE-03, FR-RPT-01 | UC-03

US-03: View Statements and Approve High-Cost Repairs (Property Owner / Landlord)

As a property owner, **I want to** view income, expenses and occupancy for my properties and approve high-cost repairs, **so that** I can control spending and evaluate the performance of my portfolio.

Acceptance Criteria

1. **Given** I own properties managed on PropRent, **when** I open my dashboard, **then** I see only my properties with monthly income, expenses, net income and occupancy rate.
2. **Given** I select a date range, **when** I export the statement, **then** a PDF or CSV containing line-level transactions is generated within 10 seconds.
3. **Given** a work order's estimated cost exceeds my approval threshold, **when** the manager assigns it, **then** it moves to Pending Owner Approval and cannot start until I approve it.

Traces to: FR-RPT-02, FR-MAINT-03, FR-AUTH-03 | UC-03 (owner approval of renewal terms)

US-04: Manage Users and Roles (Platform Administrator)

As a platform administrator, **I want to** invite users and assign least-privilege roles, **so that** every person only accesses the data they are authorised to see.

Acceptance Criteria

1. **Given** I am an administrator, **when** I invite a user and assign a role, **then** the user receives a time-limited (72-hour) activation link and cannot log in until activation.
2. **Given** I change a user's role, **when** I save, **then** the new permissions apply on that user's next request and the change is written to the audit log with my identity and timestamp.
3. **Given** I deactivate a user, **when** the action completes, **then** all active sessions and refresh tokens for that user are revoked immediately.

Traces to: FR-AUTH-01, FR-AUTH-02, FR-AUTH-03

US-05: Search and Enquire About Listings (Prospective Tenant / Applicant)

As a prospective tenant, **I want to** search available units by location, price, bedrooms and amenities and send an enquiry, **so that** I can shortlist suitable homes without calling each agent.

Acceptance Criteria

1. **Given** published listings exist, **when** I apply filters, **then** only matching Available units are shown, with results returned in under 2 seconds.
2. **Given** I view a listing, **when** I submit the enquiry form with valid contact details, **then** the manager is notified, I receive an acknowledgement and the conversation continues in a message thread.
3. **Given** a unit becomes occupied, **when** its status changes, **then** it is removed from public search within 1 minute.

Traces to: FR-PROP-03, FR-AUTH-01, FR-COMM-02

US-06: Update Job Status and Record Final Cost (Maintenance Contractor / Vendor)

As a maintenance contractor, **I want to** view assigned jobs, update progress with completion photos and record my final cost, **so that** work is clearly documented and I am paid promptly.

Acceptance Criteria

1. **Given** a manager has triaged a ticket and assigned it to me, **when** I open the job, **then** I see the description, tenant-provided media, access instructions and preferred time slots, but no unrelated tenant data.
2. **Given** I finish the job, **when** I mark it Completed with at least one photo and the final cost, **then** the manager is notified for verification.
3. **Given** the manager verifies completion, **when** they approve, **then** the work order is Closed and my final cost is recorded as approved for payment.

Traces to: FR-MAINT-02, FR-MAINT-03, FR-AUTH-03, FR-COMM-01

US-07: Process Payments Securely (Payment Gateway Provider)

As a payment gateway provider, **I want to** receive idempotent, tokenised payment requests and have my webhooks verified securely, **so that** transactions are processed without duplicate charges or compliance breaches.

Acceptance Criteria

1. **Given** a tenant pays an invoice, **when** PropRent sends the payment request, **then** it carries an idempotency key tied to the invoice ID, and card data is captured only through gateway-hosted fields (PropRent stores only the token and last four digits).
2. **Given** a webhook arrives, **when** PropRent processes it, **then** it verifies the signature and timestamp, ignores duplicate event IDs and reconciles the pending transaction without posting to the ledger twice.
3. **Given** the gateway does not respond in time, **when** the timeout occurs, **then** the transaction is marked Pending and reconciled by webhook or polling within 15 minutes with no duplicate charge.

Traces to: FR-FIN-02, FR-FIN-03 | UC-02 | NFR-SEC-03

US-08: Verify Auditability and a Bounded Baseline (Regulatory & Legal Compliance Bodies / Academic Evaluation Committee)

As a compliance reviewer or academic evaluation committee member, **I want to** inspect auditable records and a bounded, traceable requirements baseline, **so that** legal compliance and project scope can be verified.

Acceptance Criteria

1. **Given** sensitive actions occur (login, role change, lease signature, payment), **when** they complete, **then** the system records user ID, timestamp and IP address in an append-only audit log.
2. **Given** a lease has been signed, **when** anyone attempts to modify it, **then** the system rejects the change and retains the signed PDF with its audit certificate.
3. **Given** the baseline is reviewed, **when** I inspect any requirement, **then** it has a unique ID, a MoSCoW priority or measurable target, and a link to a user story or use case.

Traces to: FR-AUTH-03, FR-LEASE-02, FR-LEASE-03 | NFR-SEC-01, NFR-SEC-02, NFR-SEC-03

Story Coverage Summary

Story	Stakeholder	Functional Requirements Covered	Use Case
US-01	Tenant / Renter	FIN-01, FIN-02, MAINT-01, MAINT-02, COMM-01, RPT-03	UC-01, UC-02
US-02	Property Manager	PROP-01, PROP-02, LEASE-01, LEASE-02, LEASE-03, RPT-01	UC-03
US-03	Property Owner	RPT-02, MAINT-03, AUTH-03	UC-03
US-04	Platform Administrator	AUTH-01, AUTH-02, AUTH-03	-
US-05	Prospective Tenant	PROP-03, AUTH-01, COMM-02	-
US-06	Maintenance Contractor	MAINT-02, MAINT-03, COMM-01	UC-01
US-07	Payment Gateway Provider	FIN-02, FIN-03	UC-02
US-08	Regulatory / Academic Committee	AUTH-03, LEASE-02, LEASE-03	-

2.2 Detailed Use Cases

The three use cases below were selected because they are the highest-frequency, highest-risk interactions of the platform and exercise its principal integrations (file storage, payment gateway and e-signature).

UC-01: Submit a Maintenance Request with Photo/Video

Use Case ID & Name	UC-01: Submit Maintenance Request
Primary Actor	Tenant
Supporting Actors	Property Manager, Notification Service, Object Storage Service
Related Stories / Requirements	US-01, US-06 | FR-MAINT-01, FR-MAINT-02, FR-COMM-01
Trigger	The tenant discovers a defect or fault in the rented unit.
Preconditions	1. The tenant is authenticated and holds an Active lease for a unit. 2. The tenant's device has network connectivity.
Main Success Scenario	1. The tenant selects **New Maintenance Request**; the system shows the form pre-filled with the tenant's unit and contact details. 2. The tenant selects a category, enters a description, chooses an urgency level (Emergency, High, Normal, Low) and preferred access time slots. 3. The tenant attaches up to five photos or videos. 4. The system validates file type (JPEG, PNG, HEIC, MP4, MOV) and size (images up to 10 MB, video up to 50 MB), scans each file for malware and uploads it to object storage. 5. The tenant reviews and submits the request. 6. The system creates a ticket with a unique ID (e.g., MR-2026-004812), sets its status to Submitted and records a timestamp. 7. The system notifies the property manager through in-app and email channels (flagged high priority for Emergency requests). 8. The system displays a confirmation with the ticket ID.

Alternative / Exception Flows

ID	Condition	System Response
A1	Tenant selects Emergency (step 2)	System shows safety guidance (e.g., shut off water or gas, call emergency services) before submission and flags the ticket for immediate triage.
E1	Unsupported file type or size exceeded (step 4)	System rejects the file, shows the allowed formats and limits, and preserves all other form data.
E2	Malware detected in an attachment (step 4)	System discards the file, informs the tenant and logs a security event.
E3	Network interruption during upload (step 4)	System resumes the chunked upload when connectivity returns; after 3 failed retries the request is saved as a draft with a retry prompt.
E4	Lease is not Active (precondition failure)	System denies access to the form and directs the tenant to contact management.

Postconditions

- **Success:** A ticket exists with status Submitted, attachments are stored and linked, the manager has been notified and the event is recorded in the audit log.
- **Failure:** No partial ticket is visible to the manager; any draft remains accessible only to the tenant.

UC-02: Online Rent Payment & Automated Invoicing

Use Case ID & Name	UC-02: Online Rent Payment and Automated Invoicing
Primary Actor	Tenant
Supporting Actors	Payment Gateway, Scheduler (system actor), Property Manager, Notification Service
Related Stories / Requirements	US-01, US-07 | FR-FIN-01, FR-FIN-02, FR-FIN-03
Trigger	An invoice is generated for the billing period and the tenant chooses to pay (or autopay executes on the due date).
Preconditions	1. The tenant holds an Active lease with a defined rent amount, billing day and grace period. 2. The payment gateway integration is configured and reachable. 3. The scheduler job for invoice generation has executed for the billing period.
Main Success Scenario	1. The scheduler generates an invoice 10 days before the due date containing rent, recurring charges and any carried-forward balance, and sets its status to Due. 2. The system notifies the tenant that the invoice is available. 3. The tenant opens the invoice and selects **Pay Now**. 4. The system displays the amount due and the tenant's saved (tokenised) payment methods, or an option to add a new one. 5. The tenant selects a method and confirms. 6. The system sends a payment request to the gateway with an idempotency key tied to the invoice ID. 7. The gateway authenticates the transaction (including 3-D Secure where required) and returns an approval. 8. The system records the transaction, marks the invoice Paid and posts the amount to the property's ledger. 9. The system generates a PDF receipt, stores it in the tenant's document library, emails it to the tenant and updates the manager and owner dashboards.

Alternative / Exception Flows

ID	Condition	System Response
A1	Tenant has autopay enabled	At 06:00 on the due date the scheduler executes steps 6 to 9 without tenant interaction and sends a payment-confirmation notification.
A2	Tenant adds a new card or bank account (step 4)	Card data is captured directly by gateway-hosted fields; PropRent stores only the token and last four digits.
E1	Payment declined (step 7)	Invoice stays Due; the tenant sees the decline reason category and is offered another method; the failed attempt is logged.
E2	Gateway timeout or no response (step 6)	System marks the transaction Pending, avoids duplicate charges via the idempotency key and reconciles the final status by webhook or polling within 15 minutes.
E3	Payment not received after the grace period	System applies the configured late fee, sets the invoice Overdue, notifies the tenant and alerts the manager.
E4	Autopay fails	System notifies the tenant immediately and retries once after 48 hours before escalating.
E5	Duplicate webhook received	System detects the duplicate by event ID and ignores it without altering the ledger.

Postconditions

- **Success:** Invoice is Paid, ledger and transaction records are consistent with the gateway, the receipt is stored and sent, and dashboards reflect the payment.
- **Failure:** No ledger entry is created for unsuccessful charges; the invoice remains Due or Overdue with a complete attempt history.

UC-03: Lease Renewal & Electronic Signature

Use Case ID & Name	UC-03: Lease Renewal and Electronic Signature
Primary Actor	Property Manager
Secondary Actors	Tenant, Property Owner, E-Signature Module, Scheduler
Related Stories / Requirements	US-02, US-03 | FR-LEASE-01, FR-LEASE-02, FR-LEASE-03
Trigger	A lease reaches the configured renewal window (default 90 days before expiry).
Preconditions	1. An Active lease exists that is approaching expiry. 2. A renewal lease template is configured for the property. 3. All signers have verified email addresses on file.
Main Success Scenario	1. The scheduler identifies the expiring lease and creates a Renewal Task on the manager's dashboard. 2. The manager reviews the current terms and payment and maintenance history. 3. The manager proposes renewal terms (new rent, term length, changed clauses) and optionally requests owner approval. 4. The owner approves the proposed terms where required. 5. The system generates the renewal lease from the template, populated with the agreed terms, with a PDF preview. 6. The manager clicks **Send for Signature**; the system sends each signer a secure, unique signing link. 7. The tenant reviews the document and either signs or requests changes through the portal. 8. On completion the system stores the signed PDF and audit certificate, sets the new lease to Active from the renewal start date and updates recurring invoicing to the new rent. 9. The system notifies all parties and archives the previous lease as Expired (Renewed).

Alternative / Exception Flows

ID	Condition	System Response
A1	Tenant requests changes (step 7)	System records the comment, returns the task to the manager and versions the draft; steps 3 to 6 repeat.
A2	Tenant declines to renew	System records the decision and marks the unit as Upcoming Vacancy for marketing.
A3	Owner rejects the proposed terms (step 4)	Manager is notified with the owner's comments and revises the proposal.
E1	Signer does not sign within 7 days	System sends reminders at days 3 and 7 and alerts the manager; after 14 days the envelope expires and the task reopens.
E2	E-signature module unavailable (step 6)	System retries with exponential back-off, informs the manager and allows manual upload of a wet-signed copy with a documented justification.

Postconditions

- **Success:** A signed renewal lease is stored and immutable, billing reflects the new terms and all actions are captured in the audit trail.
- **Failure:** The existing lease remains unchanged; draft renewal documents stay in a Pending state with full version history.

3. Functional Requirements

3.1 Conventions

- Identifiers follow the pattern **FR-<MODULE>-<NN>**, where the module codes are AUTH, PROP, LEASE, FIN, MAINT, COMM and RPT.
- Priority uses the MoSCoW framework: **Must Have** (essential for the minimum viable release), **Should Have** (important, but a workaround exists), **Could Have** (desirable if time permits) and **Won't Have** (explicitly out of scope for this release).
- The word **shall** denotes a mandatory behaviour of the system.
- The **Related Story** column links each requirement to the user stories in Section 2.1.

> **Scope note**
> The baseline contains exactly **20 functional requirements** (3 each for AUTH, PROP, LEASE, FIN, MAINT and RPT; 2 for COMM). Each requirement consolidates closely related behaviour and is essential to the three core use cases or to platform security. Features removed from the earlier 108-requirement draft are recorded in Section 3.9 as deferred (Won't Have) items and are not counted in this baseline.

3.2 Epic 1: Authentication, RBAC & Profile Management (AUTH)

Secure access for six roles, with least-privilege enforcement and traceability.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-AUTH-01	**User registration & invitation onboarding.** The system shall allow tenants and prospective tenants to self-register with email and password (email verification required before first login), and shall create managers, owners, contractors and administrators only by invitation with a time-limited (72-hour) activation link.	Must Have	US-04, US-05
FR-AUTH-02	**Secure authentication & session control.** The system shall authenticate users with credentials and issue short-lived access tokens (JWT) with rotating, revocable refresh tokens; lock an account for 15 minutes after 5 consecutive failed logins; support password reset through a single-use, time-limited link; and revoke all sessions on deactivation.	Must Have	US-04
FR-AUTH-03	**RBAC, data scope & audit logging.** The system shall enforce permissions by role (Administrator, Property Manager, Owner, Tenant, Contractor, Applicant) and by data scope on every API endpoint and UI route, let administrators assign, change and revoke roles with effect on the user's next request, and record authentication events, role changes and sensitive actions with user ID, timestamp and IP address.	Must Have	US-03, US-04, US-06, US-08

3.3 Epic 2: Property & Unit Listing Management (PROP)

Publishing accurate, searchable inventory to prospective tenants.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-PROP-01	**Property & unit management.** Managers shall create, view, edit and archive properties and units (address, type, owner assignment, size, bedrooms, rent, security deposit) with a controlled unit status (Draft, Available, Reserved, Occupied, Under Maintenance, Archived) and validated transitions.	Must Have	US-02
FR-PROP-02	**Listing publication & media.** Managers shall upload up to 20 photos per unit and publish or unpublish units to the public catalogue after mandatory-field validation; each publication records who published and when.	Must Have	US-02
FR-PROP-03	**Public search & enquiry.** Visitors shall search and filter published units by location, price range, bedrooms and amenities, view a listing page with media gallery and availability date, and submit an enquiry; occupied units shall be removed from public search within 1 minute.	Must Have	US-05

3.4 Epic 3: Tenant Onboarding, Lease Agreements & Document Management (LEASE)

From approved applicant to signed, renewable and tamper-proof lease.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-LEASE-01	**Lease template & generation.** Managers shall maintain lease templates with merge fields (tenant, unit, rent, dates, deposit, clauses) and generate a lease document from a template and the agreed terms, with a PDF preview before sending.	Must Have	US-02
FR-LEASE-02	**Electronic signature.** The system shall collect legally binding e-signatures from all signers in a configurable order, send automated reminders at day 3 and day 7, expire envelopes after 14 days, and store the signed document with its audit certificate.	Must Have	US-02, US-08
FR-LEASE-03	**Lease records, renewal & document library.** The system shall store lease terms and manage lease status (Draft, Pending Signature, Active, Expired, Terminated, Renewed), create renewal tasks at a configurable window (default 90 days before expiry), and keep lease documents in a role-restricted library in which signed documents are immutable.	Must Have	US-02, US-08

3.5 Epic 4: Financial Engine, Rent Billing & Payment Processing (FIN)

Accurate billing and secure, duplicate-free rent collection.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-FIN-01	**Recurring invoice generation.** A scheduler shall generate itemised rent invoices (rent, recurring charges and carried-forward balance) per active lease at a configurable lead time (default 10 days before the due date), idempotently.	Must Have	US-01
FR-FIN-02	**Online payment, tokenisation & receipts.** Tenants shall pay invoices online by card, bank debit or digital wallet through gateway-hosted fields; the system shall store only gateway tokens and the last four digits, support saved payment methods, use idempotency keys on every payment request, and generate, store and email a PDF receipt for each successful payment.	Must Have	US-01, US-07
FR-FIN-03	**Autopay, late fees & webhook reconciliation.** Tenants shall be able to enrol in or cancel autopay with a recorded mandate (one retry after 48 hours on failure); the system shall apply configurable late fees to invoices overdue after the grace period and process gateway webhooks securely (signature verification, replay protection, duplicate-event detection) to reconcile pending transactions.	Should Have	US-01, US-07

3.6 Epic 5: Maintenance & Work Order Ticketing System (MAINT)

Structured, trackable repairs from tenant report to verified completion.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-MAINT-01	**Request submission with media.** Tenants with an active lease shall submit requests with category, description, urgency (Emergency, High, Normal, Low) and preferred access times, attaching up to 5 photos or videos (images up to 10 MB, video up to 50 MB) validated by type and scanned for malware.	Must Have	US-01
FR-MAINT-02	**Ticket tracking, triage & emergency handling.** Each request shall receive a unique ID and a time-stamped status history (Submitted, Assigned, In Progress, Pending Approval, Completed, Closed, Cancelled); emergency requests shall display safety guidance and raise a high-priority alert to the manager; managers shall prioritise tickets against SLA targets, and breaches shall be flagged on the dashboard.	Must Have	US-01, US-06
FR-MAINT-03	**Work orders, contractor updates & approvals.** Managers shall convert tickets to work orders and assign them to contractors, who can accept or decline, update status and upload completion photos and final cost; work orders above a configurable cost threshold shall require owner approval before work starts, and managers shall verify completion.	Must Have	US-03, US-06

3.7 Epic 6: Notifications, Messaging & Communication Hub (COMM)

Keeping tenants, managers and contractors informed and in contact.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-COMM-01	**In-app & email notifications.** The system shall provide an in-app notification inbox (read/unread state, deep links to the record) and send templated transactional emails for key events (invoice, payment, ticket updates, lease actions), honouring user channel preferences while legally or safety-critical notices remain mandatory.	Must Have	US-01, US-06
FR-COMM-02	**Messaging & announcements.** The system shall provide secure, threaded messaging between tenants or applicants and managers (and between managers and contractors), allow managers to broadcast announcements to all tenants of a property or building, and retain messages in a searchable archive.	Should Have	US-05

3.8 Epic 7: Reporting, Analytics & Manager Dashboard (RPT)

Decision-ready visibility for managers, owners and tenants.

Requirement ID	Feature Name / Description	Priority	Related Story
FR-RPT-01	**Manager dashboard & operational reports.** The system shall display portfolio KPIs (occupancy rate, rent collected versus expected, overdue balances, open tickets, upcoming lease expiries) and generate a rent roll and an arrears aging report (1-30, 31-60, 61-90, 90+ days), filterable by property and date range and exportable to PDF and CSV.	Must Have	US-02
FR-RPT-02	**Owner statements & dashboard.** The system shall generate periodic income and expense statements and an occupancy view for owners, limited to their own properties and exportable to PDF and CSV within 10 seconds.	Must Have	US-03
FR-RPT-03	**Tenant dashboard.** Tenants shall see their balance, next due date, payment history, active tickets and lease summary.	Could Have	US-01

3.9 Requirements Summary

Epic	Module Code	Must	Should	Could	Won't	Total
Authentication, RBAC & Profile	AUTH	3	0	0	0	3
Property & Unit Listing	PROP	3	0	0	0	3
Tenant Onboarding, Lease & Documents	LEASE	3	0	0	0	3
Financial Engine & Payments	FIN	2	1	0	0	3
Maintenance & Work Orders	MAINT	3	0	0	0	3
Notifications & Communication	COMM	1	1	0	0	2
Reporting & Analytics	RPT	2	0	1	0	3
Total		**17**	**2**	**1**	**0**	**20**

Baseline verification: 8 stakeholders | 8 user stories | 20 functional requirements | 10 non-functional requirements | **30 requirements in total** (20 FR + 10 NFR).

Deferred items (Won't Have, not counted in the baseline): social login and enterprise SSO, syndication to third-party portals, automated tenant screening, native accounting-system integration, IoT sensor-triggered tickets, automatic message translation, predictive analytics, SMS and push notifications, and bulk CSV import. These are documented to prevent scope creep and may be reconsidered in a future release.

4. Non-Functional Requirements

4.1 Conventions

Each NFR is stated with a measurable target, the rationale that justifies it and the method that verifies it. Unless stated otherwise, performance targets are measured at the 95th percentile (p95) under the reference load defined in NFR-PERF-03, in a production-equivalent environment.

> **Verification principle**
> Every NFR must be verifiable by a defined method: load testing, penetration testing, automated accessibility audit, usability testing, failover drill or static analysis. Verification methods are detailed in the test plan.

4.2 Performance & Scalability

NFR ID	Category	Requirement / Metric / Target Value	Rationale
NFR-PERF-01	Performance	Page load: Largest Contentful Paint (LCP) is below 2.5 s on a simulated 4G connection for the dashboard and listing pages. *Verification: Lighthouse / WebPageTest runs.*	Fast first render reduces abandonment of listing pages and improves the daily user experience.
NFR-PERF-02	Performance	API response time: p95 latency is below 500 ms for standard read and write endpoints, excluding third-party calls, under the reference load. *Verification: automated load testing.*	Responsive interactions are essential for task completion and perceived quality.
NFR-PERF-03	Scalability	Concurrent user capacity: the system shall support 2,000 concurrent authenticated users (10,000 registered users) while meeting NFR-PERF-01 and NFR-PERF-02, with an error rate below 1%. *Verification: load and stress testing.*	Peaks occur around rent due dates and the start of the month.

4.3 Security & Data Privacy

NFR ID	Category	Requirement / Metric / Target Value	Rationale
NFR-SEC-01	Encryption in transit and at rest	All traffic uses TLS 1.3 with HSTS enabled and weak ciphers disabled; databases, object storage and backups are encrypted at rest with AES-256, with keys managed in a key-management service. *Verification: TLS configuration scan and infrastructure audit.*	Protects credentials, personal and financial data from interception and storage compromise.
NFR-SEC-02	Access control	Authorisation is checked server-side on every request using RBAC and object-level ownership checks (e.g., an owner opening another owner's property by URL receives HTTP 403 and the attempt is logged); 100% of endpoints are covered by authorisation tests. *Verification: automated authorisation tests and penetration testing.*	Prevents broken access control and insecure direct object references.
NFR-SEC-03	Payment security (PCI-DSS)	Card data never touches PropRent servers: it is captured by gateway-hosted fields and tokenised; only the token and last four digits are stored. This keeps PropRent within PCI-DSS SAQ A scope, with a PCI-DSS Level 1 certified gateway. *Verification: SAQ A self-assessment and data-flow review.*	Minimises compliance scope and liability.

4.4 Usability & Accessibility

NFR ID	Category	Requirement / Metric / Target Value	Rationale
NFR-USA-01	Responsiveness & accessibility	Mobile-first responsive design supporting viewport widths from 320 px to 2560 px, with all tenant functions fully usable on a smartphone, conforming to WCAG 2.1 Level AA (contrast of at least 4.5:1, keyboard operability, touch targets of at least 44 x 44 px). *Verification: axe / Lighthouse accessibility score of at least 90 plus manual screen-reader testing.*	Most tenants access the portal from mobile devices, and accessibility is a legal expectation.
NFR-USA-02	Task efficiency	A first-time tenant can pay rent in at most 4 clicks from login. *Verification: usability testing with at least 8 participants.*	Quantifies intuitive navigation for the most frequent task.

4.5 Reliability & Availability

NFR ID	Category	Requirement / Metric / Target Value	Rationale
NFR-REL-01	Availability, backup & disaster recovery	Monthly uptime of at least 99.9% (about 43 minutes of unplanned downtime), excluding announced maintenance windows. Automated full database backups run daily with transaction-log archiving, retained 35 days in a separate region. A documented disaster-recovery plan targets RPO of at most 15 minutes and RTO of at most 4 hours. *Verification: uptime monitoring and a restore drill at least twice a year.*	Tenants must be able to pay rent and report emergencies at any time; financial data must survive failure.

4.6 Maintainability & Portability

NFR ID	Category	Requirement / Metric / Target Value	Rationale
NFR-MNT-01	Modular, containerised architecture & API documentation	The system is a modular architecture with module boundaries aligned to the seven epics, packaged as Docker containers deployable on any major cloud provider or on-premises. All REST interfaces are documented with OpenAPI 3.0, published via Swagger UI and kept in sync with the code. *Verification: architecture review, container build in CI and contract tests against the OpenAPI definition.*	Allows independent development and testing, avoids vendor lock-in and eases integration and handover.

4.7 NFR Traceability Overview

The matrix maps each NFR to the functional epics it most affects (● = primary impact).

NFR ID	Quality Attribute	Key Driver	AUTH	PROP	LEASE	FIN	MAINT	COMM	RPT
NFR-PERF-01	Performance	Listing search and dashboards		●		●			●
NFR-PERF-02	Performance	Responsive interactions on every API	●	●	●	●	●	●	●
NFR-PERF-03	Scalability	Billing-day peaks	●	●		●			●
NFR-SEC-01	Security	Personal, financial and lease data	●	●	●	●	●	●	●
NFR-SEC-02	Security	Role and ownership boundaries	●		●	●	●	●	●
NFR-SEC-03	Security / Compliance	Card payments				●			
NFR-USA-01	Usability	Mobile-first tenants of varied abilities	●	●		●	●	●	●
NFR-USA-02	Usability	Core tenant task efficiency				●	●		●
NFR-REL-01	Reliability	Rent collection and emergency requests			●	●	●	●	
NFR-MNT-01	Maintainability	Long-term evolution and integration changes	●	●	●	●	●	●	●

Closing note

The requirements in this chapter form the baseline for the architecture, database design, test planning and traceability matrix developed in subsequent chapters. Changes to this baseline shall follow the project's formal change-control procedure.